

Independent Analytical Note

Deriving the sp^3 Bond Angle ($109^\circ 28'$) via Pure 3D Euclidean Geometry

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Abstract

In structural chemistry, molecules exhibiting sp^3 hybridization (such as CH_4 or general AB_4 type molecules) adopt a uniform tetrahedral spatial configuration. While standard textbooks present the bond angle as $\theta \approx 109.47^\circ \approx 109^\circ 28'$, this note independently reconstructs the angle step by step from fundamental 3D Euclidean geometry and trigonometry. Rather than treating the familiar tetrahedral angle as a given result, the derivation develops it from elementary geometric relationships without relying on vector dot products.

1. Geometric Model & Nomenclature

Consider a regular tetrahedron where all four vertices represent the surrounding atoms (B), and the central atom (A) lies precisely at the centroid (center of mass) of the tetrahedron.

Let:

- x : Edge length of the regular tetrahedron (side length of the equilateral triangular faces)
- h : Height of an equilateral triangular face
- r : Circumradius of a triangular face (distance from a face's center to any of its 3 vertices)
- H : Total vertical height of the tetrahedron (perpendicular distance from any vertex to the opposite base face)
- l : Distance from the central atom (centroid) to any vertex
- θ : Bond angle subtended at the central atom by any two adjacent vertices

2. Step-by-Step Derivation

2.1 Geometry of the Equilateral Triangular Face

For a triangular face with side length x , applying Pythagoras' theorem gives the height h :

$$h = \sqrt{x^2 - \left(\frac{x}{2}\right)^2} = \frac{\sqrt{3}}{2}x \quad (1)$$

The circumradius r can be determined using Brahmagupta's Theorem and Equation (1):

$$\begin{aligned} x \times x &= 2r \times h \\ \therefore r &= \frac{x}{\sqrt{3}} \end{aligned} \quad (2)$$

2.2 Height of the Tetrahedron (H)

Consider the right-angled triangle formed by the apex vertex, the centroid of the base face, and one vertex of the base face:

$$H^2 + r^2 = x^2 \quad (3)$$

Substituting $r = \frac{x}{\sqrt{3}}$ into Equation (3):

$$\begin{aligned} H^2 + \left(\frac{x}{\sqrt{3}}\right)^2 &= x^2 \implies H^2 = x^2 - \frac{x^2}{3} = \frac{2}{3}x^2 \\ \therefore H &= \sqrt{\frac{2}{3}}x \end{aligned} \quad (4)$$

2.3 Distance from Centroid to Vertices (l)

Let the central atom be at distance l from each vertex. The vertical distance from the central atom to the base face is $(H - l)$.

Applying Pythagoras' theorem to the triangle formed by the central atom, the centroid of the base face, and a base vertex:

$$l^2 = r^2 + (H - l)^2 \quad (5)$$

Expanding Equation (5) with $r = \frac{x}{\sqrt{3}}$ and $H = \sqrt{\frac{2}{3}}x$:

$$\begin{aligned} l^2 &= \left(\frac{x}{\sqrt{3}}\right)^2 + \left(\sqrt{\frac{2}{3}}x - l\right)^2 \\ l^2 &= \frac{x^2}{3} + \frac{2}{3}x^2 - 2\sqrt{\frac{2}{3}}xl + l^2 \end{aligned}$$

Simplifying terms:

$$\begin{aligned} 2\sqrt{\frac{2}{3}}xl &= x^2 \\ l &= \frac{x}{2\sqrt{\frac{2}{3}}} = \frac{\sqrt{3}}{2\sqrt{2}}x = \sqrt{\frac{3}{8}}x \end{aligned} \quad (6)$$

2.4 Determination of the Bond Angle (θ)

Consider the isosceles triangle formed by two vertices and the central atom. The base of this triangle is the tetrahedral edge x , and the two equal legs are of length l .

Applying the Law of Cosines:

$$x^2 = l^2 + l^2 - 2(l)(l)\cos\theta = 2l^2(1 - \cos\theta) \quad (7)$$

Substituting $l^2 = \frac{3}{8}x^2$:

$$x^2 = 2 \left(\frac{3}{8}x^2 \right) (1 - \cos \theta)$$

$$1 = \frac{3}{4}(1 - \cos \theta)$$

$$\frac{3}{4} \cos \theta = \frac{3}{4} - 1 = -\frac{1}{4}$$

$$\cos \theta = -\frac{1}{3}$$

Taking the arccosine:

$$\theta = \cos^{-1} \left(-\frac{1}{3} \right) \approx 109.4712^\circ \approx 109^\circ 28' \quad (8)$$

3. Conclusion

This geometric approach independently reconstructs the $109^\circ 28'$ bond angle from elementary 3D Euclidean geometry and trigonometry rather than treating the familiar tetrahedral angle as a given result. The derivation illustrates how a standard result from structural chemistry can be reached step by step from fundamental geometric relationships, highlighting the value of geometric intuition and first-principles reasoning in structural analysis and mathematical exploration.